Why It *Hurts*:

Identifying the Drivers of Negative Thoughts in Emotional Support Conversations

EMNLP2026 BUDAPEST

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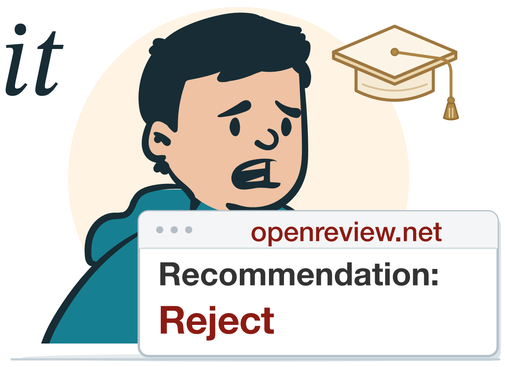
01 PROBLEM

Why does this event hurt?

Task: infer salient appraisals—the interpretations that drive distress.

*"My paper got **rejected**. I was counting on it **to graduate.**"*

"Now I don't know what would happen…"



Sixth-year CS PhD student Needs a paper to graduate Introverted · conscientious

PRE-EVENT MENTAL STATE

Belief	Desire	Intention
My research work is solid and will definitely be accepted	I will graduate this year	I will work extra hard to ensure acceptance of this paper

Illustrative appraisals	Meaning
Goal incongruence	Goal blocked
Unpredictability of consequences	Uncertain future
Unacceptable consequences	Feels unbearable

Qwen3-32B

Without PRISM

Self-cause

Peripheral appraisal

02 BENCHMARK

AppraiSal

Persona-grounded synthetic dialogues, human-annotated.

996 retained dialogues | **14** appraisal dimensions | **5** labels per dialogue

AugESC Upsetting events

PersonaHub Profiles + traits

1 Context construction

Demographics + Big Five | Subjective interpretation

2 Dialogue generation

Pseudo BDI + five salient appraisals

Help-seeker agent ⟷ Therapist agent

Exploration → Reframing → Revision

Remove pseudo appraisal labels

3 Human annotation 145 annotators

Pre-event BDI | 5 of 14 appraisals + rationales

Quality filter: mean ≥ 3/5 **1,120 → 996**

Krippendorff's α **0.776**

Agreement study: 50 entries, 3 annotators each.

03 METHOD

PRISM: Bayesian inverse planning

h = 8 in each hypothesis pool

$$\underbrace{\mathbb{P}(\mathcal{A},\mathcal{B}\mid\mathcal{D}_{1:t},\mathcal{P})}_{\text{Joint posterior}} \propto \underbrace{\mathbb{P}(\hat{u}_t\mid\mathcal{D}_{<t},\mathcal{P},\mathcal{B},\mathcal{A})}_{\text{Utterance likelihood}} \cdot \underbrace{\mathbb{P}(\mathcal{B}\mid\mathcal{D}_{<t},\mathcal{P})}_{\text{BDI posterior}} \cdot \underbrace{\mathbb{P}(\mathcal{A}\mid\mathcal{D}_{<t},\mathcal{P},\mathcal{B})}_{\text{Appraisal posterior}}$$

Inputs: Dialogue $\mathcal{D}_{1:t}$ | Persona $\mathcal{P}$ | Observed utterance $\hat{u}_t$ — BDI: belief, desire, intention

Hypothesis agent

BDI pool $\mathcal{H}^t_{\mathcal{B}}$ → Select BDI $\mathcal{B}^*_t$ → Appraisal pool $\mathcal{H}^t_{\mathcal{A}}$

Agent scoring

Token log-probabilities

Utterance agent $\mathbb{P}(\hat{u}_t\mid\cdot)$

BDI agent $\mathbb{P}(\mathcal{B}\mid\cdot)$

Appraisal agent $\mathbb{P}(\mathcal{A}\mid\cdot)$

Evidence weighting

Information gain

$$\mathrm{IG}(\pi_t) = 1 - \frac{H(\pi_t)}{\ln h}$$

From utterance likelihoods

Weight + accumulate by turn

Top-5 appraisals

Dimension scores

Flatten + softmax

Judge agent → $\hat{a}_1$ $\hat{a}_2$ $\hat{a}_3$ $\hat{a}_4$ $\hat{a}_5$

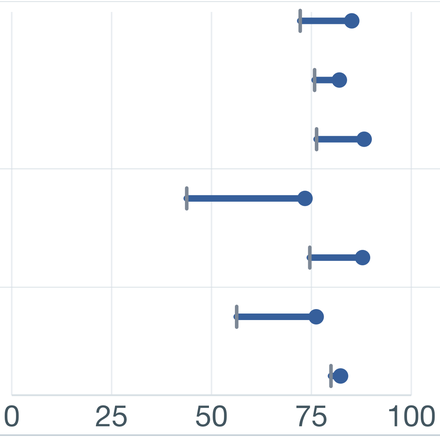
Next turn · retain · replace · rerank

04 APPRAISAL INFERENCE

PRISM improves all 7 backbones

Precision@1: top prediction in the annotated appraisal set.

Precision@1 (%) Qwen / Gemma: instruct · gpt-oss: reasoning



	Baseline	PRISM	Gain (pp)
Qwen3-4B	72.2	**85.1**	**+12.9**
Qwen3-32B	75.8	**82.0**	**+6.2**
Qwen3-30B-A3B	76.3	**88.2**	**+11.9**
Gemma3-4B	43.8	**73.4**	**+29.6**
Gemma3-27B	74.6	**87.8**	**+13.2**
gpt-oss-20B	56.3	**76.2**	**+19.9**
gpt-oss-120B	79.9	**82.3**	**+2.4**

05 REFRAMING · SEPARATE STUDY

Salience-guided reframing

Qwen3-4B appraisals → **gpt-oss-120B** responses.

Human preference (%) Guided win | Tie | Loss

	Guided win	Tie	Loss
Overall preference			
vs. Ordinary	50	12	38
vs. Exhaustive	50	16	34
Situation analysis			
vs. Ordinary	38	28	34
vs. Exhaustive	44	16	40
Mental analysis			
vs. Ordinary	52	22	26
vs. Exhaustive	54	8	38

Response-length difference controlled to **<10 words**.

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Poster made with GPT-6.1-Sol.

Data & code

github.com/seacowx/AppraiSal-and-PRISM